

Description

Farmer John built a cowshed with N ($2 \leq N \leq 100000$) compartments, which are distributed on a straight line with coordinates $x_1, \dots, x_N$ ($0 \leq x_i \leq 1000000000$). His C ($2 \leq C \leq N$) cows are not satisfied with the distribution of compartments and are angry because of the presence of other cows in the cowshed. To prevent cows from fighting each other, Farmer John wants to place these cows in designated compartments so that the minimum distance between any two adjacent cows is as large as possible. The **maximum** value of this **minimum** distance is what needs to be found.

Input

The first line of the input contains two integers N and C separated by a space. The second line contains N integers representing the coordinates of each compartment.

Output

The output should be a single integer representing the maximum minimum distance between adjacent cows.

Sample Input

```
5 3
1
2
8
4
9
```

Sample Output

```
3
```